Exp No: 02

Date: 27/7/26

Study of different types of Network cables.

A1 Understand different types of Network Cable.

Different types of cables used in networking are

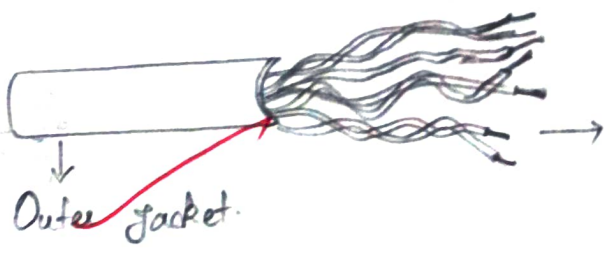
1. Unshielded Twisted pair (UTP)
2. Shielded Twisted pair (STP)
3. Coaxial Cable
4. Fibre Optic Cable.

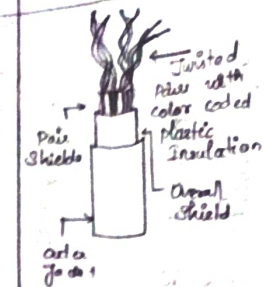
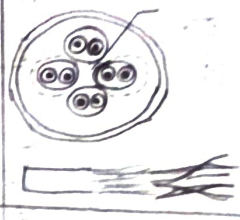
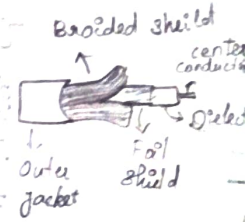
1. Cable Type:

UTP

Category:	Maximum Data Transmission	Advantage/ Disadvantage	Application/ Use
Category 3	10 bps	Advantage: cheeper in cost Easy to install as they a smaller Overall diameter. Disadvantage: More prone to (EMI) Electromagne -tic interference & noise	10 Base-T Ethernet
Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet
Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet

Image:



Cable Type	Category	Maximum Data Transmission	Advantages/ Disadvantage	Application/use	Image
STP	category 6, 6a	10 Gbps	Advantages Shielded Faster than UTP less susceptible to noise & interference Disadvantages Expensive Greater installation effort	Gigabit Ethernet, 10G Ethernet (55m) widely used in Data center.	
	category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m).	
Coaxial Cable	RG-6 RG 59 RG11	10-100 Mbps	High bandwidth Immune to Interference Low loss bandwidth Versatile	Speed of signal is 500m Television network High speed internet connections.	
	Single Mode Multi Mode	100 Gbps	Disadvantages: Limited distance cost Size is bulky Advantages: High Speed High Bandwidth High Security Disadvantage Expensive Requires skilled installers	Maximum distance of fibre Optics cable is around 100 meters	

b) Make Your Own ~~Tool~~ Ethernet cross-Over Cable / Straight cable.

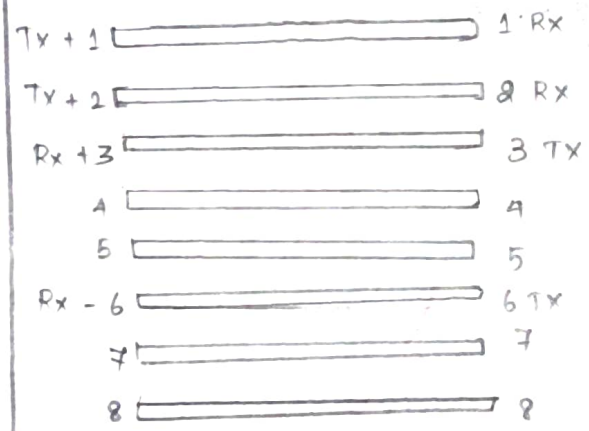
Tools and parts needed:

Ethernet cabling: CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.

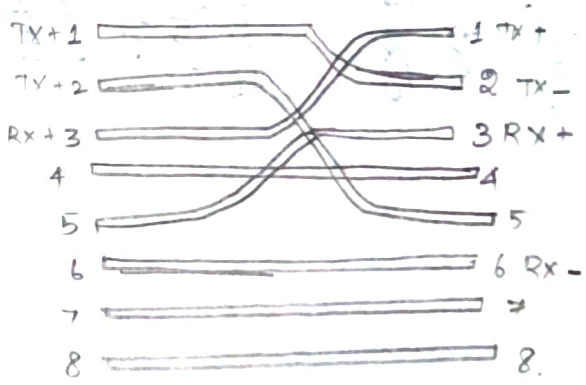
A crimping tool: This is an all-in-one networking tool. Shaped to push down the pins in the plug and strip and cut the shielding off the cable.

Two RJ45 plugs.
Optional two plug shields.

Straight thru cable



X-Over cable.



Step:

1. To start construction of the device, begin by threading shields onto the cable.

2. Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

3. After, you will need to untangle the wires, there should be four "Twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A & the other in B

4. Once the Order is correct, bunch them together in a line, & if there are any that stick out further than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the Order.

5. Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if isn't, that means that you stripped off too much shielding.

6. After the wires are securely sitting inside the plug, insert it into the crimping tool and push down

Lastly, repeat for the other end using diagram B.

Result:

Different types of cable Network are learned & Execute Successfully.